

Practice Questions (50 SQL Questions)

Basic Queries

1. Show all employees.
 2. Show only employee names and salaries.
 3. Find employees from Ahmedabad.
 4. Find employees with salary > 60000.
 5. Find female employees.
 6. Find employees between age 25 and 30.
 7. Sort employees by salary descending.
 8. Find top 5 highest salaries.
 9. Count total employees.
 10. Find minimum and maximum salary.
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WHERE + LIKE + IN

11. Find employees whose name starts with 'A'.
 12. Find employees whose city ends with 'a'.
 13. Find employees in Ahmedabad, Surat, Pune.
 14. Find employees not in IT department.
 15. Find employees with salary between 50000 and 70000.
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Aggregate Functions

16. Find average salary.
 17. Find total salary department-wise.
 18. Find average age city-wise.
 19. Count employees in each city.
 20. Find highest salary in each department.
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GROUP BY + HAVING

21. Find departments with more than 3 employees.
 22. Find cities where average salary > 60000.
 23. Find departments with total salary > 200000.
 24. Find cities having more than 2 employees.
 25. Find average salary by gender.
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JOINS

- 26. Show employee name with department name.
 - 27. Show all employees and their project names.
 - 28. Find employees working in IT.
 - 29. Show project budgets with department names.
 - 30. Find employees whose department budget > 300000.
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Subqueries

- 31. Find second highest salary.
 - 32. Find employees earning above average salary.
 - 33. Find highest paid employee in each department.
 - 34. Find departments with no employees.
 - 35. Find employees hired before the oldest IT employee.
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Window Functions

- 36. Rank employees by salary.
 - 37. Dense rank employees department-wise.
 - 38. Find running total of salaries.
 - 39. Find top 3 salaries in each department.
 - 40. Find previous employee salary using `LAG()`.
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Date Functions

- 41. Find employees hired after 2021.
 - 42. Find employees hired in January.
 - 43. Find years of experience.
 - 44. Find orders placed in March.
 - 45. Find monthly sales amount.
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Advanced Queries

- 46. Create a view for high salary employees.
- 47. Create stored procedure for department employees.
- 48. Use transaction with `COMMIT` and `ROLLBACK`.
- 49. Create index on salary column.
- 50. Find duplicate cities in employee table.

Bonus Hard Questions

- Find nth highest salary
- Find employees earning more than department average
- Pivot city-wise employee count
- Recursive CTE example
- Find gaps in employee IDs
- Department-wise top performer
- Running balance calculation

